

AI-Powered Hospital Management System

Software Requirements Specification (SRS)

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1. Introduction

The purpose of this project is to develop an AI-Powered Hospital Management System that helps hospitals manage patients, doctors, appointments, medical records, billing, and healthcare activities efficiently. The system integrates artificial intelligence to provide a smart symptom checker and assist medical staff in delivering better patient care.

Scope

The system will provide:

- Patient Management
- Doctor Management
- Appointment Management
- Medical Records Management
- Billing and Invoice Management
- AI Symptom Checker
- Prescription Management
- Report Generation

2. Users of the System

Admin

- Manage patients and doctors
- Manage hospital departments
- Generate reports
- Publish announcements

Doctor

- View and manage appointments

- Write prescriptions
- Update patient medical records
- View patient history

Patient

- Register and manage profile
- Book appointments
- View medical records
- Use AI symptom checker
- View billing and invoices

Receptionist

- Register new patients
- Schedule appointments
- Manage billing

3. Functional Requirements

FR-01 Login System

Users can log in using their credentials based on their role.

FR-02 Patient Management

Admin and receptionist can add, update, delete, and search patients.

FR-03 Doctor Management

Admin can add, update, and delete doctor profiles.

FR-04 Appointment Management

Patients can book appointments; doctors and receptionists can manage schedules.

FR-05 Medical Records Management

Doctors can create and update patient medical records; patients can view their own records.

FR-06 Billing Management

Receptionist can generate invoices; patients can view and pay bills.

FR-07 Prescription Management

Doctors can write and manage prescriptions for patients.

FR-08 AI Symptom Checker

Patients can input symptoms and receive AI-generated preliminary health advice.

FR-09 Report Generation

Admin can generate hospital performance and patient statistics reports.

4. Non-Functional Requirements

Performance

The system should respond within 3 seconds under normal load.

Security

Only authorized users can access the system; all sensitive data must be encrypted.

Reliability

The system should maintain 99.9% uptime to support critical hospital operations.

Usability

The interface should be simple, intuitive, and accessible for all staff levels.

Scalability

The system should handle increasing numbers of patients and doctors without performance degradation.

5. Use Cases

Patient

- Login
- Book Appointment
- View Medical Records
- Use AI Symptom Checker
- View Billing

Doctor

- Login

- Manage Appointments
- Write Prescription
- Update Medical Records

Receptionist

- Login
- Register Patient
- Schedule Appointment
- Generate Invoice

Admin

- Login
- Manage Patients
- Manage Doctors
- Manage Departments
- Generate Reports

6. Future Scope

- Mobile Application for Android and iOS
- Telemedicine and Video Consultation
- AI-Based Disease Prediction
- Integration with Pharmacy Systems
- Lab Test Management Module

7. Conclusion

The AI-Powered Hospital Management System will modernize hospital operations by automating administrative tasks and providing intelligent healthcare support through AI technology. The system aims to improve patient outcomes, reduce waiting times, and enable data-driven decision-making across all hospital departments.